Day 16 Revision Sheet - Wed, Nov 4, 2026

1. Key ideas

- **KPI** = a metric tied to a goal. Name, formula, period, target, owner. A bare name (best selling product) is not a KPI. Prefer ratio KPIs: add-to-cart rate, return rate, repeat purchase rate.
- **Leading vs lagging** is always relative to an outcome. Name the outcome first: visits lead orders; orders lead revenue; revenue lags both. Units sold records sales after they happen, so it is lagging.
- **Metric vs dimension**: a metric can be added, averaged or counted; a dimension is grouped or filtered by. Some fields are both: age bands (dimension) vs average age (metric); delivery days bucketed (dimension); return flag counted gives returned orders (metric); returned / orders = return rate (KPI). Unit price and discount % are non-additive: use a weighted average.
- **Vanity metric**: looks big, drives no decision (page views, followers, downloads: cumulative totals only rise and say nothing about current orders). Repeat purchase rate and AOV are not vanity: they tie to money and fall when business is bad. AOV = total revenue / number of orders.

2. The profit case: orders up, profit down

- Profit = Revenue - Cost. Cost = Revenue - Profit (revenue and profit given, cost derived). Margin % = Profit / Revenue x 100.
- First number to check: profit per order and margin %. It shows whether the problem is revenue per order or cost per order, so it decides the next check.
- Volume effect = extra orders x OLD profit per order. Margin effect = NEW orders x change in profit per order. Volume + margin = total change in profit.
- Answer structure: Finding, Hypotheses (labelled as guesses), Data to confirm, Action with a target.

Worked example (all values checked by running the calculation):

	Last month	This month
Orders / AOV	1,000 / Rs 500	1,200 / Rs 500
Revenue = orders x AOV	Rs 500,000	Rs 600,000
Profit (given)	Rs 100,000	Rs 80,000
Cost = revenue - profit	Rs 400,000	Rs 520,000
Cost per order	Rs 400	Rs 433.33
Profit per order	Rs 100	Rs 66.67
Margin	20%	13.33%

Volume effect: 200 x Rs 100 = +Rs 20,000. Margin effect: 1,200 x (-Rs 33.33) = -Rs 40,000. Net: 100,000 + 20,000 - 40,000 = Rs 80,000 (matches). Revenue per order did not change; cost per order rose. The problem is on the cost side.

Candidate causes (hypotheses, need a cost breakdown): product cost, shipping, packaging, ad cost per order, returns, shift to low-margin products. A discount lowers price per order, so it cannot raise cost per order while AOV is unchanged.

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Mistakes, improvement, practice to retry, recall, open items

3. My mistakes and the fix

What I got wrong	Fix
'Best selling product' as a KPI; units sold as leading	A KPI needs a number, target and period. Units sold records sales after they happen: lagging.
Called orders leading, then lagging	Name the outcome first. Orders lead revenue but lag visits.
10 fields: labels only, never asked 'could it be both?'	Test each field: group by it, or add/average/count it? Many can be both.
Skipped AOV and repeat purchase rate in the vanity question	They are not vanity: they tie to money and behaviour.
Profit case: AOV and discount were the only checks	Add margin, cost per order, profit per order. Profit has a revenue side and a cost side.
Discount named as a cause of higher cost per order	A discount cuts price per order. Cost per order rises from shipping, packaging, ads, product cost, returns.
Recommendation 'review costs' repeated a check	Name the data to pull, the action and a target.
Recall: profit = revenue x cost; margin = revenue / profit	Profit = revenue - cost. Margin = profit / revenue x 100.
Recall: 'volume' defined, but no effect on profit	Volume effect = extra orders x old profit per order.

4. How I improved

- Got every per-order number right with the formula written, and worked revenue and cost for both months step by step.
- Saw that an unchanged AOV can still hide a profit fall, and said it in my own words.
- Gave a correct 'both metric and dimension' example (age) in recall.

5. Practice to retry (no solutions)

- Rewrite the profit-case answer lines 4 and 5: three causes marked as hypotheses; the data to pull, the action and a target.
- New case: last month 2,000 orders, AOV Rs 400, profit Rs 160,000. This month 2,400 orders, AOV Rs 400, profit Rs 120,000. Find margin and profit per order for both months, then volume effect and margin effect, and check they add up to the change.
- Rewrite the 5 KPIs with at least two ratio KPIs. For each: name, formula, period, target, owner, and leading or lagging against a named outcome.
- Pick 3 of the 10 order fields and show how each can act as both a dimension and a metric.

6. Recall questions (answer from memory)

- Profit, cost and margin: write each formula without looking.
- Which number do you check first when orders rise and profit falls, and why?
- What do the volume effect and the margin effect measure? Give the formula for each.
- Name the five parts of a KPI. Is 'best selling product' one?
- Leading or lagging: units sold, visits, orders? Against which outcome?
- Why are page views a vanity metric, and why is AOV not?

7. Open items

- Profit-case retest (lines 4 and 5) in a future session.
- Planned for Nov 4 but not done: Functions lesson, is_prime, average(list), grade counts, AI skill hour, Git conflict redo from a blank folder.
- Watch list: write formulas in the right order (profit, margin); name the outcome before leading/lagging.